

Lab Report 4: UART Driver and Serial Communication

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Abstract—This report documents the design and bare-metal implementation of a UART communication firmware for the NUCLEO-F411RE development board (STM32F411RE, ARM Cortex-M4). The project configures USART2 to transmit serial data at 115200 baud, enabling real-time telemetry output to a VS Code Serial Monitor. A modular driver architecture separates low-level register access (uart_driver, gpio_driver) from application-level logic (serial, utils, sensor). ADC readings from a potentiometer are formatted as human-readable strings and transmitted every 500 ms using a TIM5-based blocking delay.

Index Terms—STM32F411RE, USART2, UART, bare-metal, CMSIS, BRR, CR1, SR, DR, serial communication, ADC, NUCLEO-F411RE.

I. INTRODUCTION

Serial communication via UART is one of the most fundamental debugging and telemetry tools in embedded systems development. Unlike hardware signals verified with an oscilloscope, UART allows the microcontroller to transmit structured, human-readable data to a host computer in real time, enabling runtime monitoring of sensor values, state variables, and system events.

This laboratory practice implements a complete UART transmission pipeline on the STM32F411RE using a bare-metal C architecture. USART2, mapped to the ST-Link virtual COM port on the NUCLEO-F411RE, is configured at 115200 baud to transmit formatted telemetry strings. A utils module handles integer-to-ASCII conversion without the standard library, a serial module provides a serial_printf-style API, and an adc_driver module reads a potentiometer connected to PA1. Every 500 ms a formatted message is transmitted over USART2:

```
"ADC Value: %d | Voltage: %d mV\r\n"
```

The practice reinforces the layered driver philosophy established in previous labs: each module

encapsulates a single hardware concern, exposing a clean API to the layer above and depending only on modules below it.

II. MATERIALS AND EQUIPMENT

The following hardware and software resources were used:

- NUCLEO-F411RE board (STM32F411RE, Cortex-M4)
- 10 kΩ potentiometer connected to PA1 (ADC1_IN1) with 3.3 V supply
- USB-A to Mini-USB cable (power and ST-LINK programming/VCP)
- Personal computer, Windows 11
- Visual Studio Code with Serial Monitor extension
- MSYS2 / MinGW64 terminal, arm-none-eabi-gcc cross-compiler
- OpenOCD with STM32CubeIDE ST-LINK scripts (flashing)
- Custom Makefile build system
- CMSIS STM32F411xE device headers (stm32f411xe.h)

III. DEVELOPMENT

A. Project Structure and Layered Architecture

The firmware follows a strict layered architecture that separates hardware access from application logic. Fig. 1 illustrates the dependency graph. At the lowest level, gpio_driver manages all GPIO register operations, including the alternate-function configuration required to route USART2 signals to PA2 (TX) and PA3 (RX). The uart_driver abstracts every USART2 peripheral register — BRR, CR1, SR, and DR — exposing a clean two-function API: uart_init() and uart_write(). The utils module provides all string formatting and integer-to-ASCII conversion without any standard library dependency. The serial module ties these

together into a developer-friendly interface, while sensor, led, and timer complete the application layer.

B. GPIO Driver

The `gpio_driver` was extended from previous labs to support USART alternate function configuration. For USART2 on the NUCLEO-F411RE, PA2 (TX) and PA3 (RX) are configured as Alternate Function 7 (AF7) by writing to `GPIOA->AFR[0]` and setting `GPIOA->MODER` bits to 10b (alternate function mode). Both pins are set to push-pull output type and high speed. The `gpio_set_af()` function isolates the AFR index and shift computation from the caller, preventing common bit-manipulation errors when crossing the `AFR[0]/AFR[1]` boundary at pin 8.

C. UART Driver

The `uart_driver` provides a low-level API for USART2. `uart_init()` performs the following sequence: (1) enable GPIOA clock and configure PA2/PA3 as AF7; (2) enable the USART2 APB1 clock via `RCC->APB1ENR |= RCC_APB1ENR_USART2EN`; (3) write BRR with the pre-calculated divisor; (4) set CR1 to enable the transmitter (TE), receiver (RE), and the USART itself (UE).

The baud rate register value is computed from the reference manual formula for 16× oversampling:

$$\text{USARTDIV} = f_{\text{PCLK1}} / (16 \times \text{Baud}) = 16,000,000 / 1,843,200 = 8.680$$

$$\text{DIV_Mantissa} = 8, \text{DIV_Fraction} = \text{round}(0.680 \times 16) = 11$$

$$\text{BRR} = (8 \ll 4) | 11 = 0x008B = 139$$

The `uart_write()` function polls `SR.TXE` until the transmit data register is empty, then writes the byte to DR. Reading DR is not required to clear TXE in transmit-only mode; TXE is automatically cleared by hardware when DR is written and is re-set when DR is transferred to the shift register. The TC (transmission complete) flag is not used here because back-to-back character transmission is handled safely by the TXE polling loop.

D. Utils Module

The `utils` module provides two key functions. `utils_itoa()` converts a signed or unsigned 32-bit integer to an ASCII string in any base from 2 to 16. It accumulates digits in reverse order into a local buffer by computing `digit = num % base` iteratively, then calls `utils_memReverse()` before null-terminating. `utils_snprintf()` parses a format string character by character, dispatching to `utils_itoa()` for `%d/%u/%x` specifiers and to `utils_printString()` for `%s`. This design eliminates any dependency on `stdio.h`, `stdlib.h`, or heap allocation.

E. Serial Module

The serial module wraps `uart_driver` and `utils` into a developer-friendly API. `serial_init()` calls `uart_init(USART_BAUD_115200)` internally. `serial_printf()` initializes a `va_list`, passes it directly to `utils_vsnprintf()` to format into a 128-byte stack-allocated buffer, then transmits each character with `uart_write()`. The `va_list` is passed by reference to avoid the loss of positional state that would occur if `va_arg` were called in separate function invocations.

F. Application Integration

The `main()` function calls `led_init()`, `timer_init()`, `sensor_init()`, and `serial_init()` in sequence, then enters an infinite loop. On each iteration the ADC is sampled via `sensor_start_conversion()` and `sensor_read_value()`, the raw 12-bit value is converted to millivolts using integer arithmetic ($\text{mV} = (\text{raw} \times 3300) / 4095$), and the formatted telemetry string is transmitted. LED2 on PA5 toggles as a heartbeat indicator every iteration. TIM5 enforces the 500 ms period via polling of the UIF flag.

```
// Main loop (every 500 ms)
sensor_start_conversion();
uint16_t raw = sensor_read_value();
uint32_t mv = ((uint32_t)raw * 3300U) / 4095U;
serial_printf("ADC Value: %d | Voltage: %d mV\r\n",
              (int)raw, (int)mv);
led_toggle();
timer_delay_ms(500);
```

IV. RESULTS

A. Serial Monitor Output

After flashing the firmware, the VS Code Serial Monitor was opened and configured to 115200 baud on the ST-Link virtual COM port. The device immediately began transmitting telemetry lines at the expected 500 ms interval. Table I presents a set of representative ADC readings collected while rotating the potentiometer across its full travel. The raw values span the expected 0–4095 range and the millivolt conversion tracks the 3.3 V reference linearly, confirming correct operation of both the ADC driver and the BRR-based baud rate configuration.

TABLE I

Representative ADC Telemetry Readings

Pot. Position	ADC Raw (0–4095)	Voltage (mV)	Duty Cycle (%)
Minimum (GND)	0	0 mV	0
~25 %	1024	825 mV	25
~50 %	2048	1650 mV	50
~75 %	3072	2475 mV	75
Maximum (3V3)	4095	3300 mV	100

B. UART Signal Verification

The UART TX signal on PA2 was monitored with a logic analyzer set to asynchronous serial mode at 115200 baud, 8N1. Fig. 3 shows the decoded waveform of a single transmitted character. The bit period measured at 8.68 μ s matches the theoretical value of $1/115200 \approx 8.68 \mu$ s, confirming that the BRR register was configured correctly and the APB1 clock is running at the expected 16 MHz.

C. BRR Calculation Verification

The BRR value of 139 (0x008B) was independently verified by back-calculating the actual baud rate it produces:

$$\text{Actual baud} = 16,000,000 / (16 \times 8.6875) = 115,107 \text{ baud}$$

The error relative to the target is $(115,200 - 115,107) / 115,200 = 0.08 \%$, well within the $\pm 2 \%$ tolerance required for reliable UART reception specified in RM0383 §19.3.4. No framing errors or character corruption were observed during extended testing.

D. Functional Compliance Summary

TABLE II

UART SRS Functional Requirements Compliance

FR	Requirement	Implementation	Status
FR-1	Enable USART2 APB1 clock	RCC->APB1ENR = USART2EN	✓ Pass
FR-2	Configure BRR for target baud	BRR = 139 (115200 @ 16 MHz)	✓ Pass

FR-3	Enable TE and UE in CR1	CR1 = TE RE UE	✓ Pass
FR-4	Poll SR.TXE before writing	while(!(SR & TXE));	✓ Pass
FR-5	Write 8-bit data to DR	DR = byte & 0xFF	✓ Pass

V. CONCLUSION

This laboratory practice successfully implemented a complete bare-metal UART telemetry pipeline on the STM32F411RE. The `uart_driver` was designed following the provided SRS, covering clock enablement, GPIO alternate-function routing, BRR calculation, and polled character transmission. The BRR value of 139 produces a baud rate error of 0.08 %, well within the tolerance for reliable UART communication.

The layered architecture — `gpio_driver` → `uart_driver` → `utils` → `serial` — proved effective at isolating hardware concerns from application logic. The `serial_printf()` function built on top of `utils_vsnprintf()` provides a familiar formatting interface without any standard library dependency, making it portable to any bare-metal STM32 project.

The ADC and UART subsystems operated concurrently without interference, demonstrating that the peripheral clock enable sequence and register initialization order described in RM0383 are both necessary and sufficient for correct operation at 16 MHz HSI. The potentiometer readings tracked the 3.3 V reference linearly across the full 12-bit range, and all five SRS functional requirements were verified on hardware.

Future work could extend the driver to support DMA-based transmission, eliminating the polling overhead in `uart_write()` and allowing the CPU to perform other tasks during long transmissions. Interrupt-driven reception would further enable bidirectional communication for command parsing.

Github links:

1. **Daniel:** <https://github.com/lopezdaniel-web/MSE-Lab>
2. **Paetz:** <https://github.com/JPaez725/MSE-Lab>
3. **Cristian:** <https://github.com/CrisxCross15/MSE-Lab>

4. **Luis:**<https://github.com/alugobji/MSE-Lab/tree/main/Lugo>

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